



BITS Pilani

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CE F231 Fluid Mechanics

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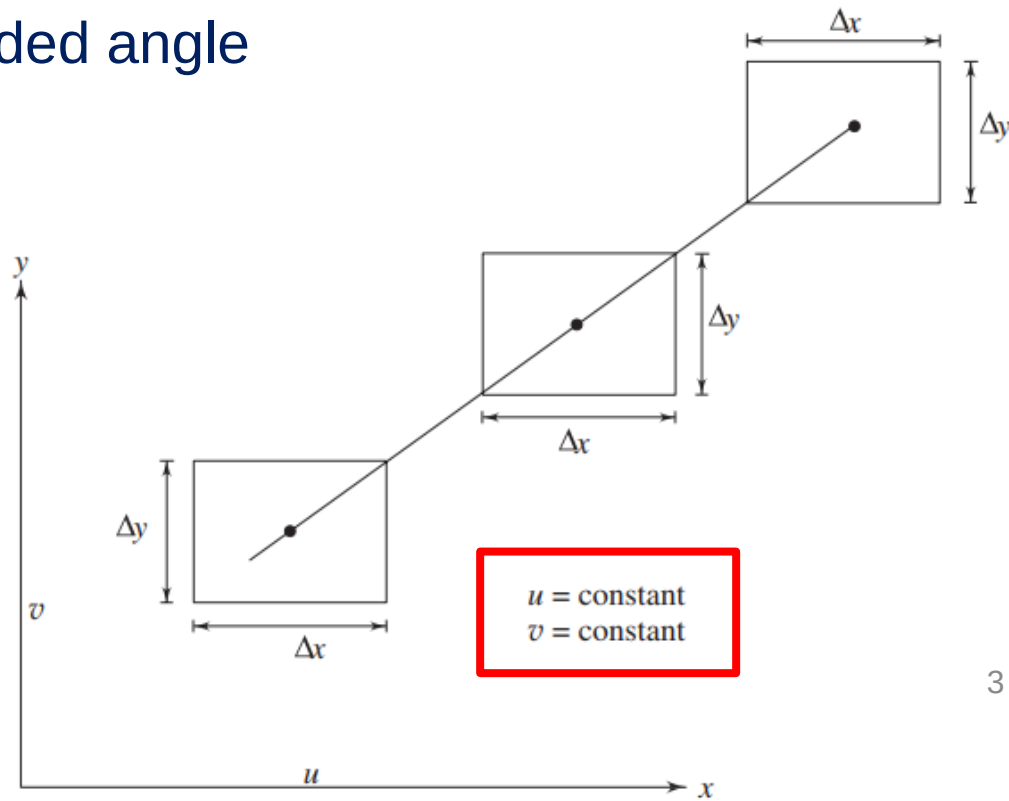


Lecture 16: Fluid Kinematics

Recap: Types of Motion

➤ Linear translation

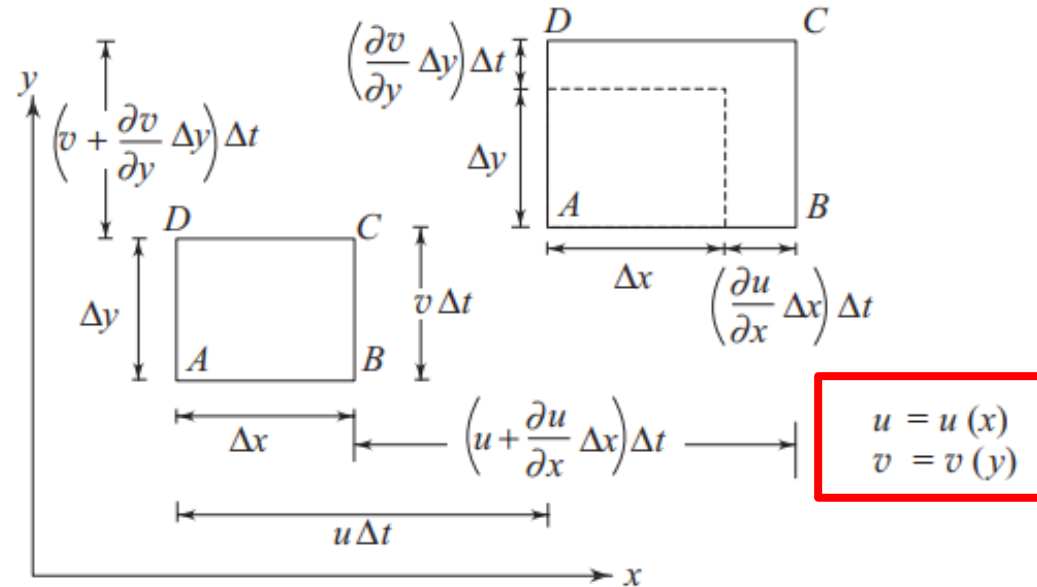
- u (x-component) and v (y-component) are not functions of x and y (space) → uniform flow
- u and v are constants
- No change in length and included angle
- Sides are displaced parallelly



Recap: Types of Motion

➤ Translation with Linear deformation

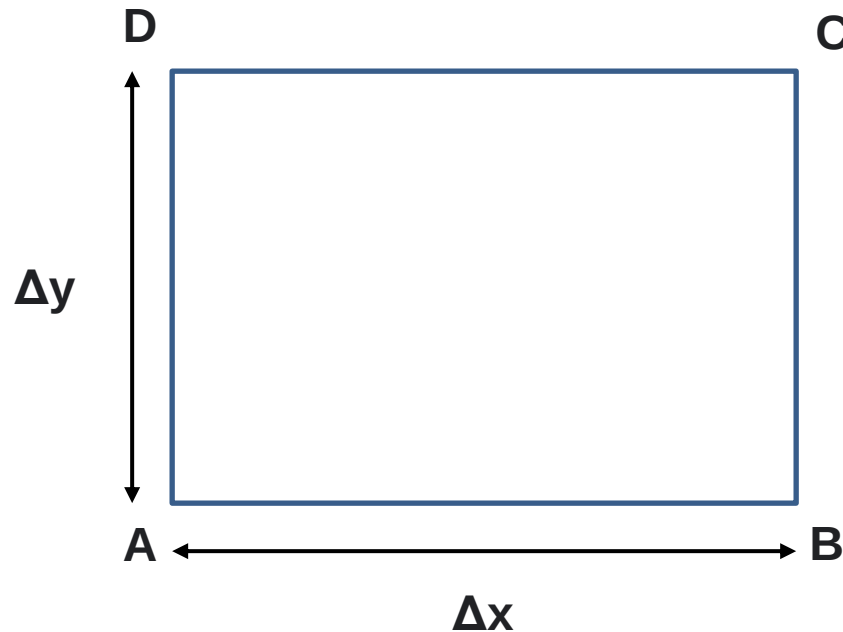
- u (x-component) and v (y-component) are functions of one space coordinate only
- u is a function of x only
- v is a function of y only
- Change in linear dimension
- No change in included angle
- Sides are displaced parallelly
- Linear strain rates



$$\dot{\epsilon}_{xx} = \frac{\partial u}{\partial x} \quad \dot{\epsilon}_{yy} = \frac{\partial v}{\partial y}$$

Types of Motion

- Translation with linear and angular deformations
 - u (x-component) and v (y-component) are functions of both space coordinates $\rightarrow u = u(x, y)$ and $v = v(x, y)$

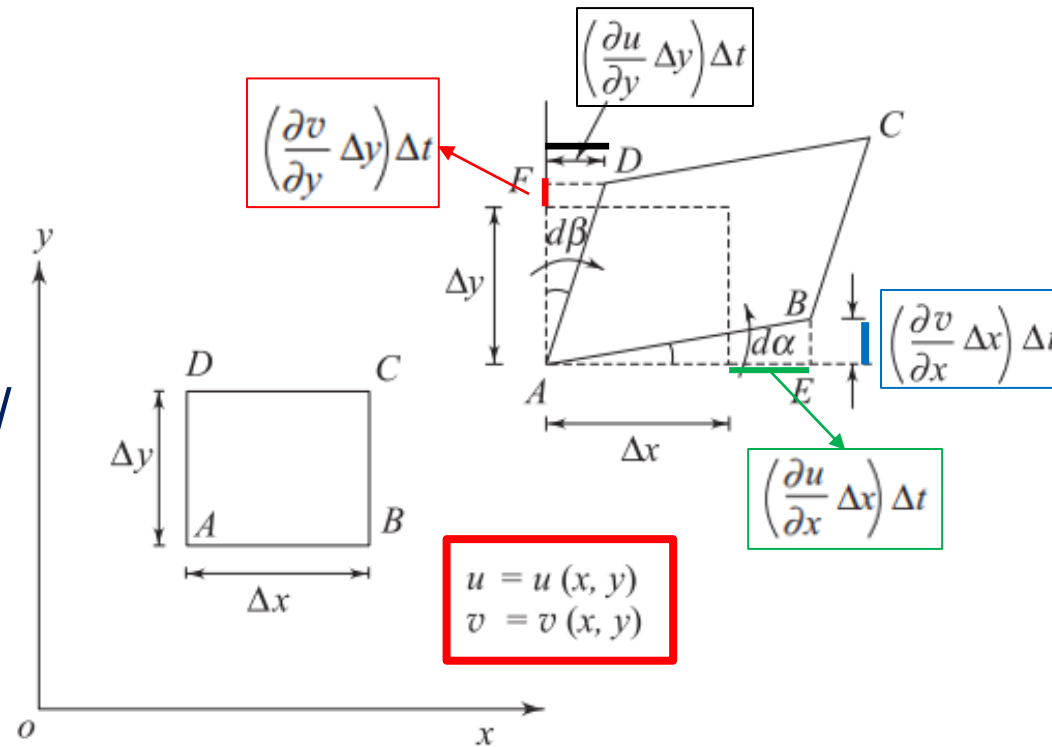


Types of Motion

➤ Translation with linear and angular deformations

- u (x-component) and v (y-component) are functions of both space coordinates
- u is a function of x and y
- v is a function of x and y
- Change in linear dimension
- Change in included angle
- Rate of angular deformation/ shear strain rate

$$\dot{\gamma}_{xy} = \left(\frac{d\alpha}{dt} + \frac{d\beta}{dt} \right)$$



Types of Motion

➤ Translation with linear and angular deformations

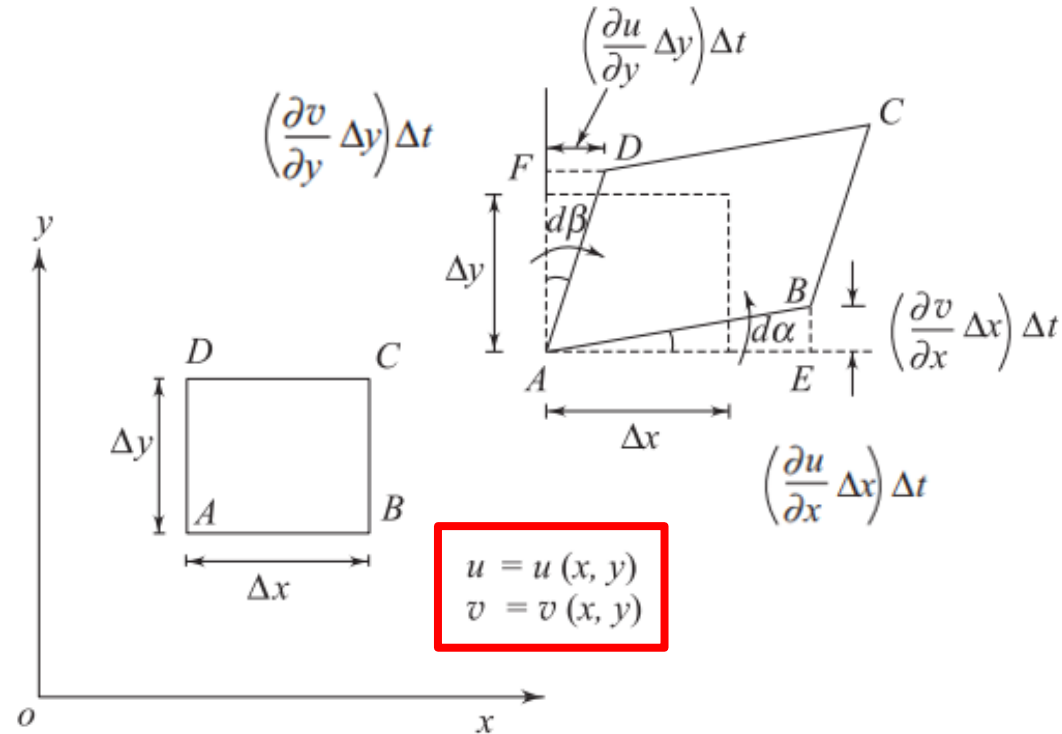
- Rate of angular deformation/shear deformation

$$\dot{\gamma}_{xy} = \left(\frac{d\alpha}{dt} + \frac{d\beta}{dt} \right)$$

$$d\alpha = \lim_{\Delta t \rightarrow 0} \tan^{-1} \left(\frac{\frac{\partial v}{\partial x} \Delta x \Delta t}{\Delta x \left(1 + \frac{\partial u}{\partial x} \Delta t \right)} \right) = \frac{\partial v}{\partial x} dt$$

$$d\beta = \lim_{\Delta t \rightarrow 0} \tan^{-1} \left(\frac{\frac{\partial u}{\partial y} \Delta y \Delta t}{\Delta y \left(1 + \frac{\partial v}{\partial y} \Delta t \right)} \right) = \frac{\partial u}{\partial y} dt$$

$$\dot{\gamma}_{xy} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}$$



Types of Motion

➤ Rotation

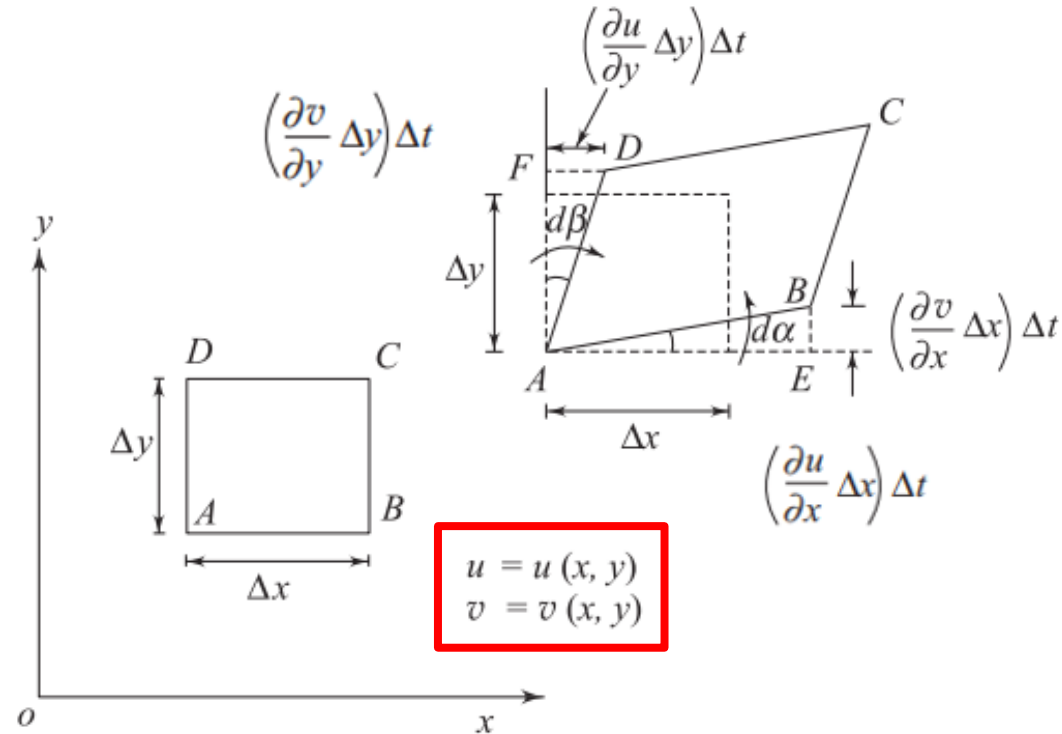
- Rotation = average angular velocity

$$\omega_z = \frac{1}{2} \left(\frac{d\alpha}{dt} - \frac{d\beta}{dt} \right)$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

$$d\alpha = \lim_{\Delta t \rightarrow 0} \tan^{-1} \left(\frac{\frac{\partial v}{\partial x} \Delta x \Delta t}{\Delta x \left(1 + \frac{\partial u}{\partial x} \Delta t \right)} \right) = \frac{\partial v}{\partial x} dt$$

$$d\beta = \lim_{\Delta t \rightarrow 0} \tan^{-1} \left(\frac{\frac{\partial u}{\partial y} \Delta y \Delta t}{\Delta y \left(1 + \frac{\partial v}{\partial y} \Delta t \right)} \right) = \frac{\partial u}{\partial y} dt$$



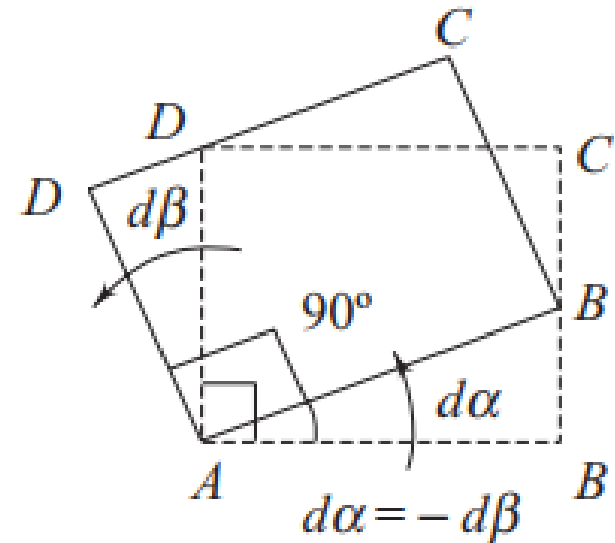
Types of Motion

➤ Rotation without angular deformation

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \quad (v_B > v_A \text{ and } u_A > u_D)$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \Rightarrow \omega_z = \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

$$\dot{\gamma}_{xy} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \Rightarrow \dot{\gamma}_{xy} = 0$$



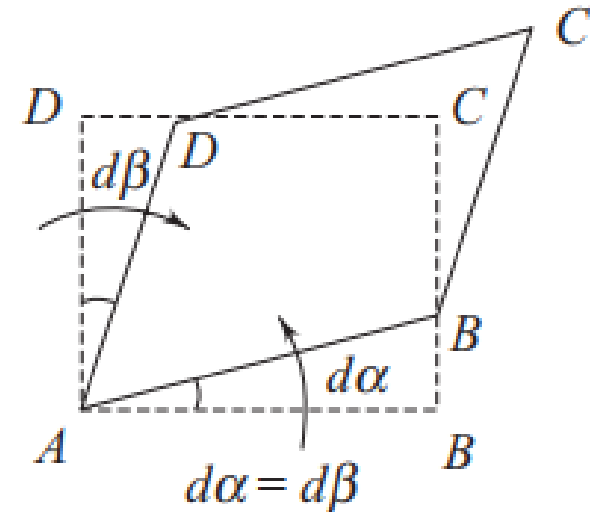
Types of Motion

➤ Angular deformation without rotation

$$\frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}$$

$$\dot{\gamma}_{xy} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \Rightarrow \dot{\gamma}_{xy} = 2 \frac{\partial v}{\partial x} = 2 \frac{\partial u}{\partial y}$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \Rightarrow \omega_z = 0$$



Types of Motion

➤ Rotation

In a three dimensional flow, the components of rotation are defined

$$\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right)$$

$$\omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right)$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

$$\vec{\omega} = \frac{1}{2} (\nabla \times \vec{V})$$

$\equiv$

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$$

(Curl)

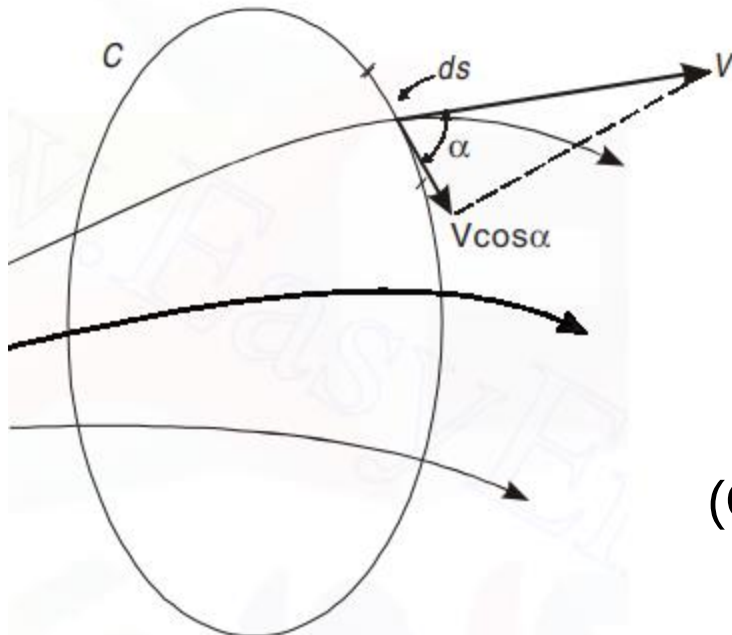
and

Necessary and sufficient condition for irrotational flow is $\nabla \times \vec{V} = 0$

Circulation and Vorticity

➤ Circulation

- The flow along a closed curve is called *circulation*
- Circulation is the line integral of the tangential component of the velocity vector, taken around a closed curve



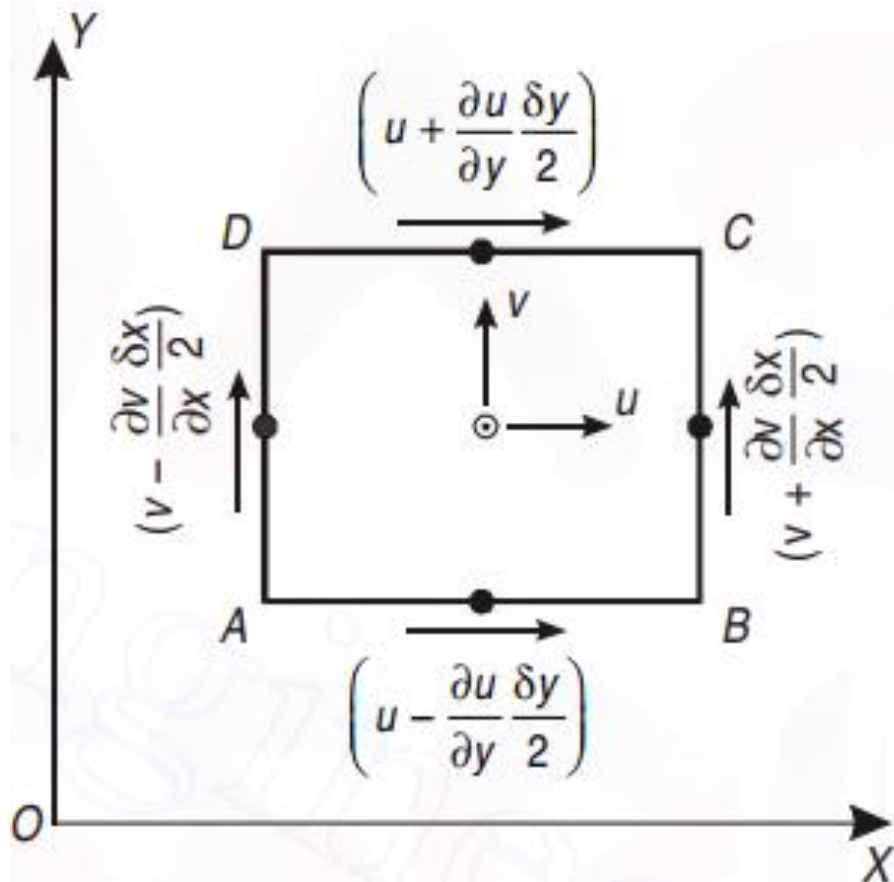
$$\Gamma = \int_C V \cos \alpha \, ds$$

$$\Gamma = \int_C (u \, dx + v \, dy + w \, dz)$$

(Cap. Gamma)

Circulation and Vorticity

➤ Circulation around elementary rectangle



$$\text{Circulation along } AB = \left(u - \frac{\partial u}{\partial y} \frac{\delta y}{2} \right) \delta x$$

$$\text{Circulation along } BC = \left(v + \frac{\partial v}{\partial x} \frac{\delta x}{2} \right) \delta y$$

$$\text{Circulation along } CD = - \left(u + \frac{\partial u}{\partial y} \frac{\delta y}{2} \right) \delta x$$

$$\text{Circulation along } DA = - \left(v - \frac{\partial v}{\partial x} \frac{\delta x}{2} \right) \delta y$$

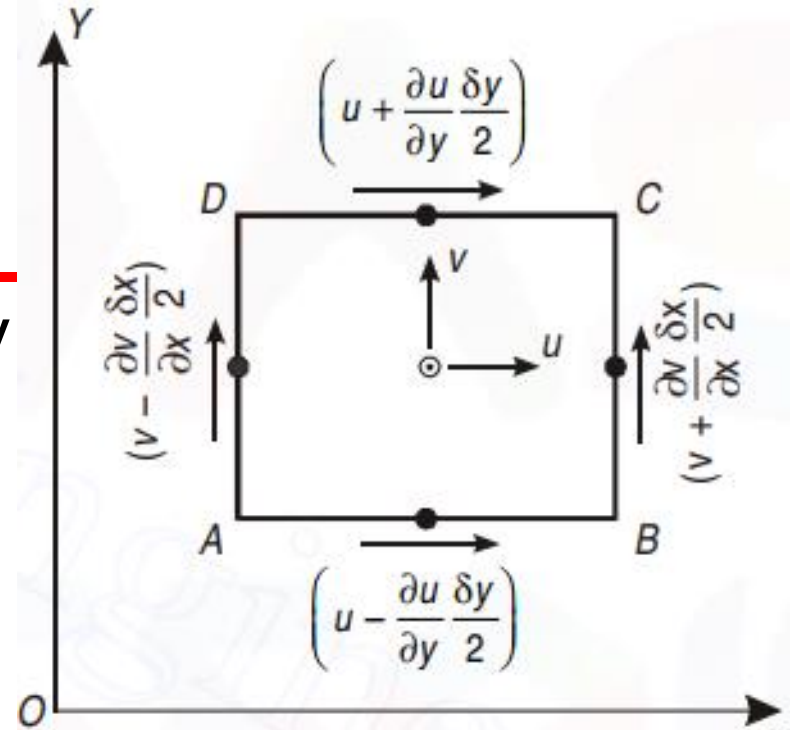
Circulation and Vorticity

- Circulation around elementary rectangle

$$\Gamma = \Gamma_{AB} + \Gamma_{BC} + \Gamma_{CD} + \Gamma_{DA}$$

$$= \left(u - \frac{\partial u}{\partial y} \frac{\delta y}{2} \right) \delta x + \left(v + \frac{\partial v}{\partial x} \frac{\delta x}{2} \right) \delta y - \left(u + \frac{\partial u}{\partial y} \frac{\delta y}{2} \right) \delta x - \left(v - \frac{\partial v}{\partial x} \frac{\delta x}{2} \right) \delta y$$

$$= \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \delta x \delta y$$



Circulation and Vorticity

➤ Vorticity

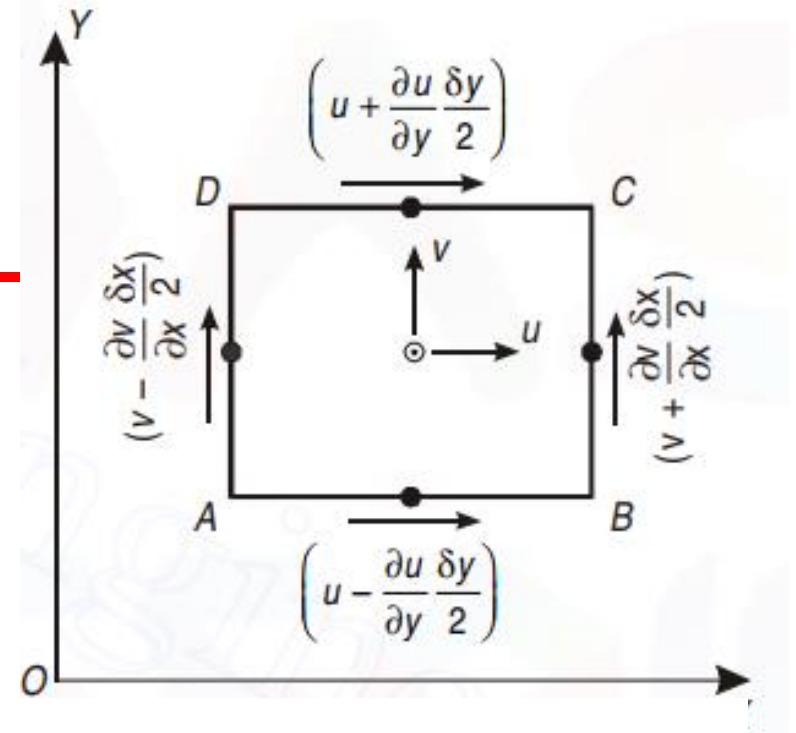
$$\zeta = \frac{\text{Circulation}}{\text{Area}}$$

(Zeta)

$$= \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

$$\zeta = 2\omega_z$$

Vorticity = 2 x Rotation



$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

Rotation, Circulation and Vorticity

➤ Vorticity = 2 x Rotation

$$\vec{\omega} = \frac{1}{2}(\nabla \times \vec{V})$$

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} = \zeta = 2\omega$$

Necessary and sufficient condition for irrotational flow is $\nabla \times \vec{V} = 0$



THANK YOU